7. Mahendra Verma, Practical Numerical Computing Using Python: Scientific & Engineering Applications, Self-Published
8. One material and tutorials: www.python.org; https://www.learnpython.org; www.vpython.org; https://www.pythonlikeyoumeanit.com.

Skills and Employability: Entire course contents with tutorials and assignments help build foundations and develops computational thinking, programming skills – design and implementation of software for scientific, engineering and industrial computing applications in universities, industries and research labs/organizations.

Evaluation Pattern: As in the rules for *Assessment Procedure* (R.14)

24MAT208 Introduction to Linear Algebra (3-1-0-4)

Course Outcomes

On successful completion of the course, students shall be able to

CO1: apply the basic arithmetic operations on vectors and matrices, including inversion and determinants, using technology where appropriate;

CO2: apply concepts and solve systems of linear equations, using technology to facilitate row reduction

CO3: apply the basic concepts of linear algebra in Euclidean spaces, including linear independence, spanning, basis, rank, nullity, subspace, and linear transformation.

CO4: apply the appropriate techniques to find eigen values and eigenvectors of a matrix or a linear transformation, and using them to diagonalize a matrix

CO5: apply the notions of Orthogonally, diagonalize symmetric matrices and quadratic forms.

Unit 1

Matrices, Operations on Matrices-Addition, Multiplication, Transpose, Special types of matrices. systems of linear equations.

Unit 2

Gaussian elimination and row operations, Echelon form of a matrix, Elementary matrices and rank of a matrix, Existence of solution of AX=B.

Unit 3

Properties of determinants, Cofactor expansion, Inverse of a matrix. Vector spaces, Subspaces.

Unit 4

Span and linear independence, Basis and dimension, Row and column space of a matrix, Change of Basis. Linear transformations, Range space and rank, null space and nullity, Matrix representation, Isomorphism.

Unit 5

Eigenvalues and Eigenvectors, Diagonalization and Similar Matrices, Real Quadratic Forms.

Textbook:

David C. Lay, *Linear Algebra and its Applications*, Pearson.

References:

Gilbert Strang, *Linear Algebra and its Applications*, Cengage.
Derek Robinson, *A Course in Linear Algebra with Applications*, Allied publishers.

24MAT209 Transforms & PDE (3-1-0-4)

Course Outcomes
On successful completion of the course, students shall be able to

CO1: apply basic concepts to obtain Fourier series of simple continuous and piece-wise periodic functions, determine Fourier transforms of simple functions.

CO2: apply basic Sturm-Liouville theory to analyse different types of orthogonal functions and determine generalized Fourier series

CO3: comprehend the mathematical model of one- and two- dimensional wave and heat Equations and apply the method of separation of variables and theory of Fourier series for rectangular geometries to solve the boundary value problems.

CO4: apply the general principle in boundary value problems for PDEs to choose coordinates that make the formula for the boundary as simple as possible

CO5: apply the principles of method of separation of variables to solve the boundary value problems in polar, cylindrical and Spherical coordinates for cylindrically symmetric problems.

Unit 1
Fourier series—Even and Odd functions, Half range expansions, Approximation by Trigonometric Polynomials. (Sections 11.1 to 11.4)

Unit 2
Sturm-Liouville problems, Generalized Fourier Series, Fourier Integral and Fourier transforms. (Sections 11.5 to 11.10)

Unit 3
Basic concepts of PDEs, Solution by Separating Variables, D'Alembert's Solution of the Wave Equation. (Sections 12.1 to 12.4)

Unit 4
Heat Equation, Solution By Fourier Series, 2D Heat Equation and Dirichlet Problem, Heat equation for long bars, Solution by Fourier Integrals and Transforms, Two Dimensional Wave Equation. (Sections 12.5 to 12.8)

Unit 5
Laplacian in Polar Coordinates, Fourier Bessel Series, Laplace's equation in Cylindrical and Spherical Coordinates, Solution of PDEs by Laplace Transforms. (Sections 12.8 to 12.12)

Textbooks
1. Erwin Kreyszig, *Advanced Engineering Mathematics*, 10th edition, Wiley.